# MANAGEMENT INFORMATION SYSTEMS
# ICT2162
# LECTURE 02

Ms. Shelomi De Alwis
Department of ICT
Faculty of Technology
University of Ruhuna

# Global E-Business

# What we discuss today?

1. Business Processes and Information Systems
2. Types of Information Systems
   - ✓ Systems for Different Management Groups
     - ✓ Transaction Processing Systems
     - ✓ Systems for Business Intelligence
       1. Management Information Systems
       2. Marketing Information System
       3. Human Resource Information System
       4. Financial Management Information System
       5. Production Management Information System
       6. Decision Support Systems
       7. Executive Support Systems

# What we discuss today?

- ✓ Systems for Linking the Enterprise
  - ✓ Enterprise applications
    1. Enterprise Systems
    2. Supply Chain Management (SCM) Systems
    3. Customer Relationship Management Systems
    4. Knowledge Management Systems (KMS)
  - ✓ Intranets and Extranets

- ✓ E-business, E-commerce and E-government

# Objectives

1. What are business processes?
2. How are they related to information systems?
3. How do systems serve the different management groups in a business?
4. How do systems that link the enterprise improve organizational performance?

# Business Processes and Information Systems

- Business processes:
  - Sets of activities,steps required to produce a product or service
  - Workflows of material,information, knowledge
  - May be tied to functional area or be cross functional

    Ex: Functional area – Human Resources – Hiring employees

    Ex: Cross Functional – Fulfilling customer order

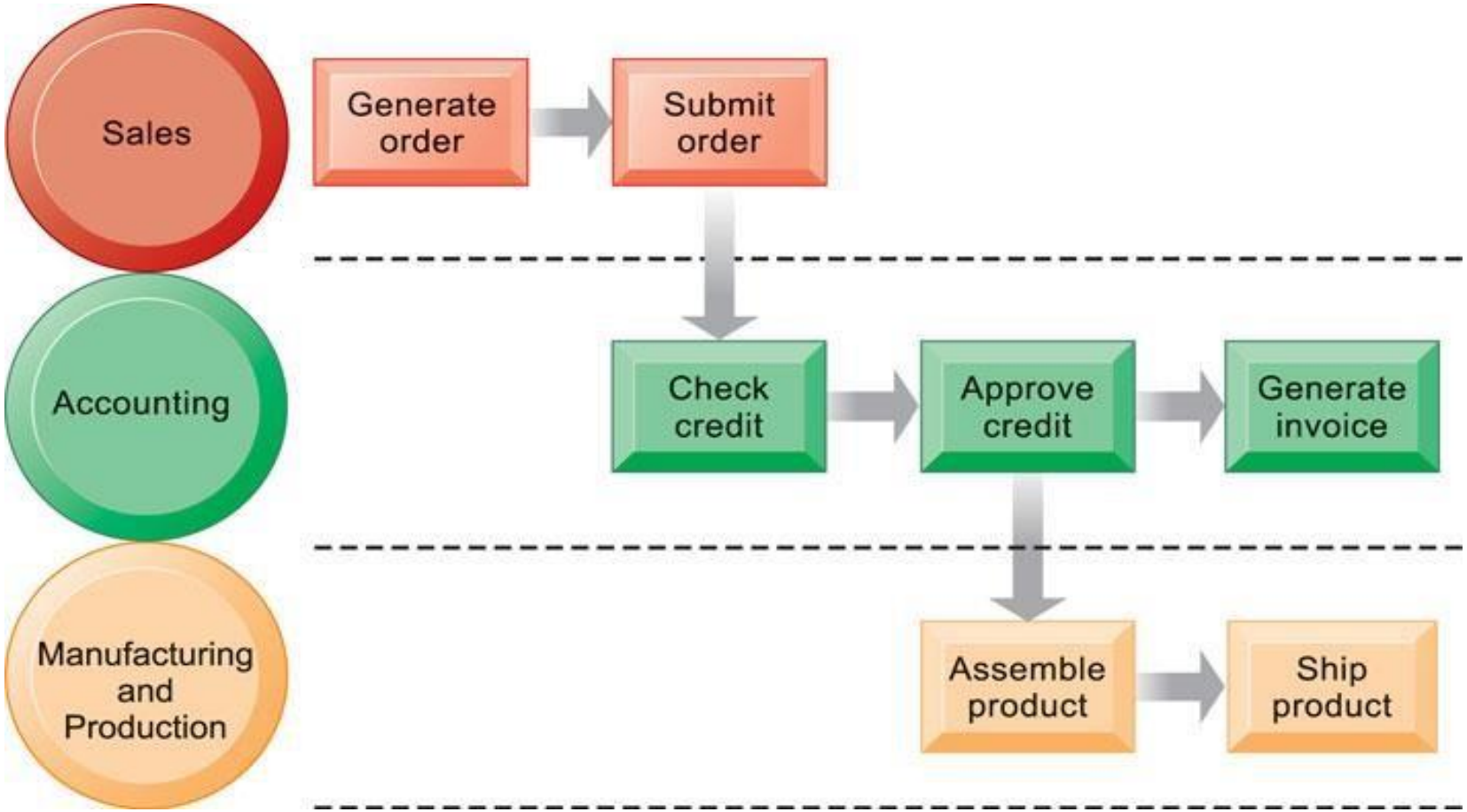
- Business processes may be assets or liabilities
- Businesses: Can be seen as collection of business processes

# Business Processes and Information Systems

**Examples of Functional Business Process**

- Manufacturing and production
  - Assembling the product, Checking for quality, Producing bills of materials
- Sales and marketing
  - Identifying customers, Making customers aware of the product, Selling the product
- Finance and accounting
  - Paying creditors, Creating financial statements, Managing cash accounts
- Human resources
  - Hiring employees, Evaluating employees' job performance, Enrolling employees in benefits plans

# Business Processes and Information Systems




The Order Fulfillment Process

# Business Processes and Information Systems

Information technology enhances business processes in two main ways

1. Increasing efficiency of existing processes
   - Automating steps that were manual
2. Enabling entirely new processes that are capable of transforming the businesses
   - Change flow of information
   - Replace sequential steps with parallel steps
   - Eliminate delays in decision making
   - Support new business models

     Ex: Kindle e-book from Amazon, buying a computer online at Best Buy

# Types of Information Systems

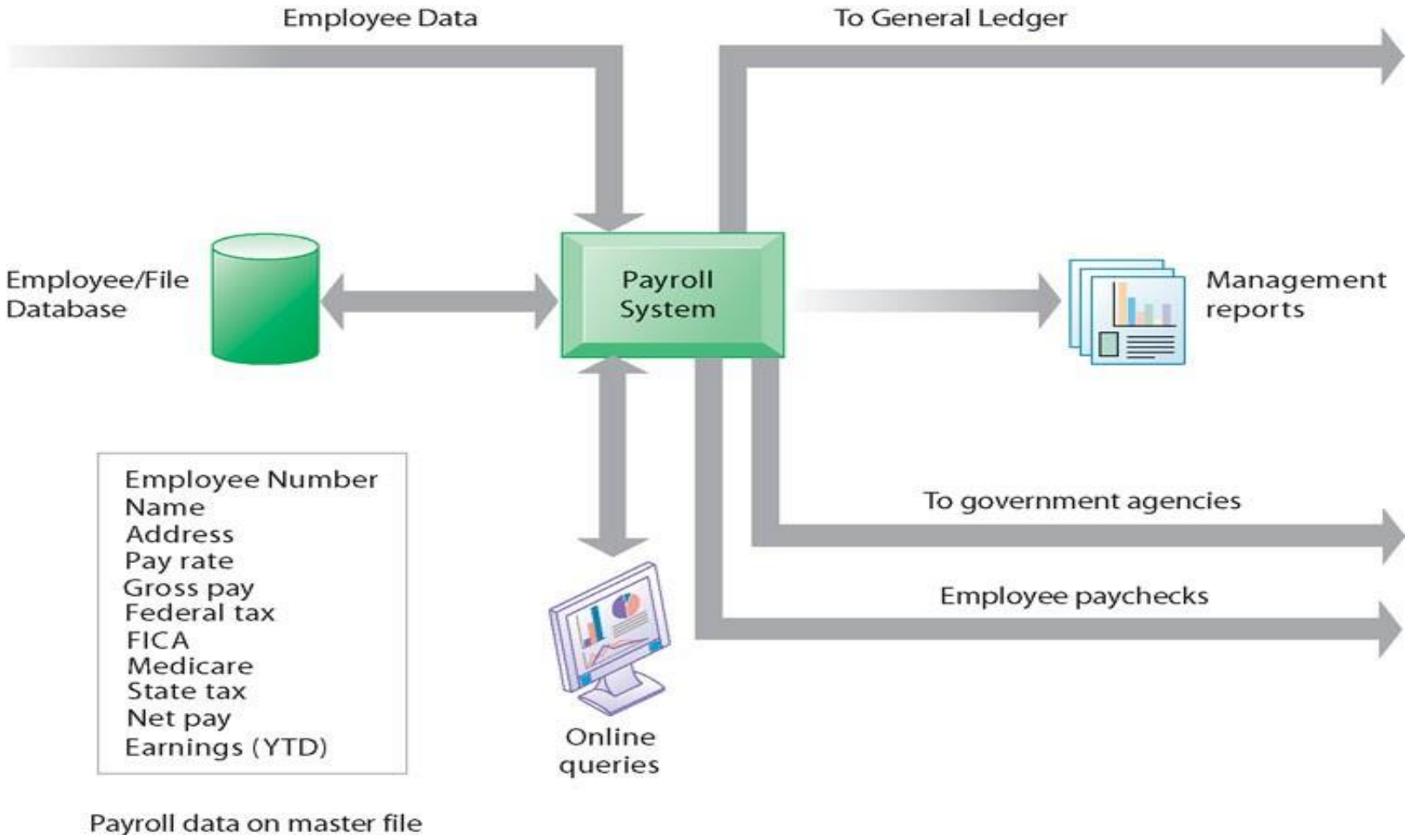
Systems for Different Management Groups

- **Transaction Processing Systems (TPS)**
- Systems for Business Intelligence

# Types of Information Systems

- **Transaction Processing Systems (TPS)**
  - Perform and record daily routine transactions necessary to conduct business
    - Examples: sales order entry, payroll, shipping
  - Allow managers to monitor status of internal operations and relations with external environment
  - Serve operational levels
  - Serve predefined, structured goals and decision making

# Types of Information Systems




A Payroll TPS - A payroll system keeps track of money paid to employees.

# Types of Information Systems

Systems for Different Management Groups

- Transaction Processing Systems
- **Systems for Business Intelligence**

# Types of Information Systems

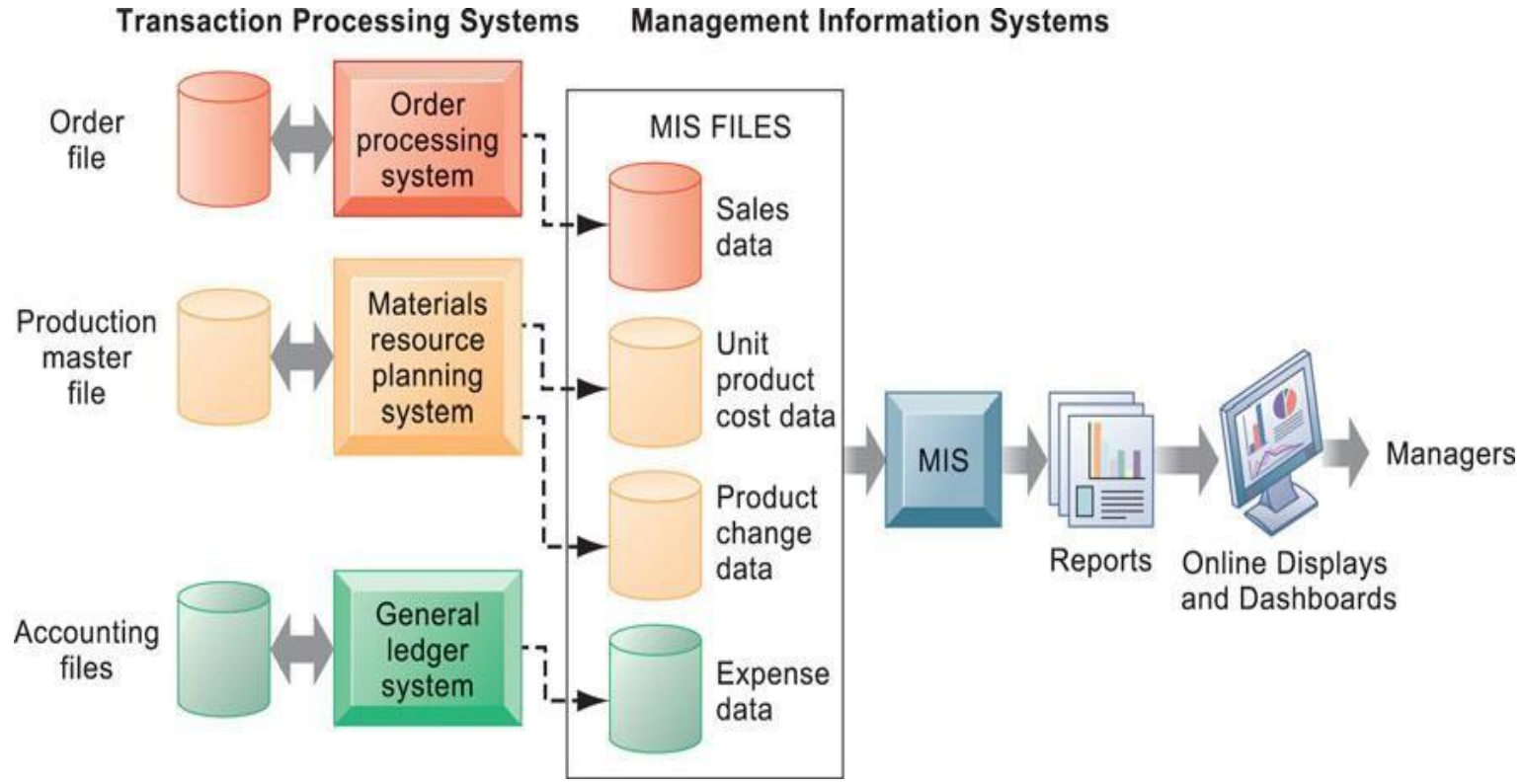
- **Business Intelligence**
  - Class of software applications
  - Analyze current and historical data to find patterns and trends and aid decision-making
  - Used in systems that support middle and senior management
  - BI for Middle Management support Monitoring, Controlling, Decision-making and Administrative activities
  - BI for Senior Management support more non routine decision making

# Types of Information Systems

## 1. Management Information Systems(MIS)

- Serve middle management
- Provide reports on firm's current performance,based on data from TPS
- Provide answers to routine questions with predefined procedure for answering them
- Typically have little analytic capability

# How Management Information Systems Obtain Their Data From the Organization's TPS




In the system illustrated by this diagram, three TPS supply summarized transaction data to the MIS reporting system at the end of the time period. Managers gain access to the organizational data through the MIS, which provides them with the appropriate reports.

# Sample MIS Report

Consolidated Consumer Products Corporation Sales by Product and Sales Region: 2011

| PRODUCT CODE | PRODUCT DESCRIPTION | SALES REGION | ACTUAL SALES | PLANNED | ACTUAL versus PLANNED |
|---|---|---|---|---|---|
| 4469 | Carpet Cleaner | Northeast | 4,066,700 | 4,800,000 | 0.85 |
| | | South | 3,778,112 | 3,750,000 | 1.01 |
| | | Midwest | 4,867,001 | 4,600,000 | 1.06 |
| | | West | 4,003,440 | 4,400,000 | 0.91 |
| | TOTAL | | 16,715,253 | 17,550,000 | 0.95 |
| 5674 | Room Freshener | Northeast | 3,676,700 | 3,900,000 | 0.94 |
| | | South | 5,608,112 | 4,700,000 | 1.19 |
| | | Midwest | 4,711,001 | 4,200,000 | 1.12 |
| | | West | 4,563,440 | 4,900,000 | 0.93 |
| | TOTAL | | 18,559,253 | 17,700,000 | 1.05 |

Summarized annual sales data, was produced by the MIS

# Types of Information Systems

## 2. Marketing Information Systems

- Help managers to identify key customers.
- Help managers with market segmentation.
  - Leads to identify target market accurately

# Types of Information Systems

## 3. Human Resource Information Systems(HRIS)

- Allow HR Managers to take better and timely decisions
- Modules
  - Payroll Module
  - Time and attendance Module
  - Benefits administration Module
  - HR Management Module
  - Training Module
  - Employee self- service Module

# Types of Information Systems

## 4. Finance Management Information Systems(FMIS)

- Allows managers to evaluate
  - the sources of funds that would be best suited for the firm,
  - those that bear the least risk,
  - and those that reaps the most gain.

- Equity and Debt

# Types of Information Systems

## 5. Production Management Information Systems(PMIS)

- Concerned with the planning and control of all activities that transform input into output by adding some value to the former to make later.
- Production planning
- Production control
- Product design
- Product development
- Plant location and layout

# Types of Information Systems

## 5. Decision Support Systems

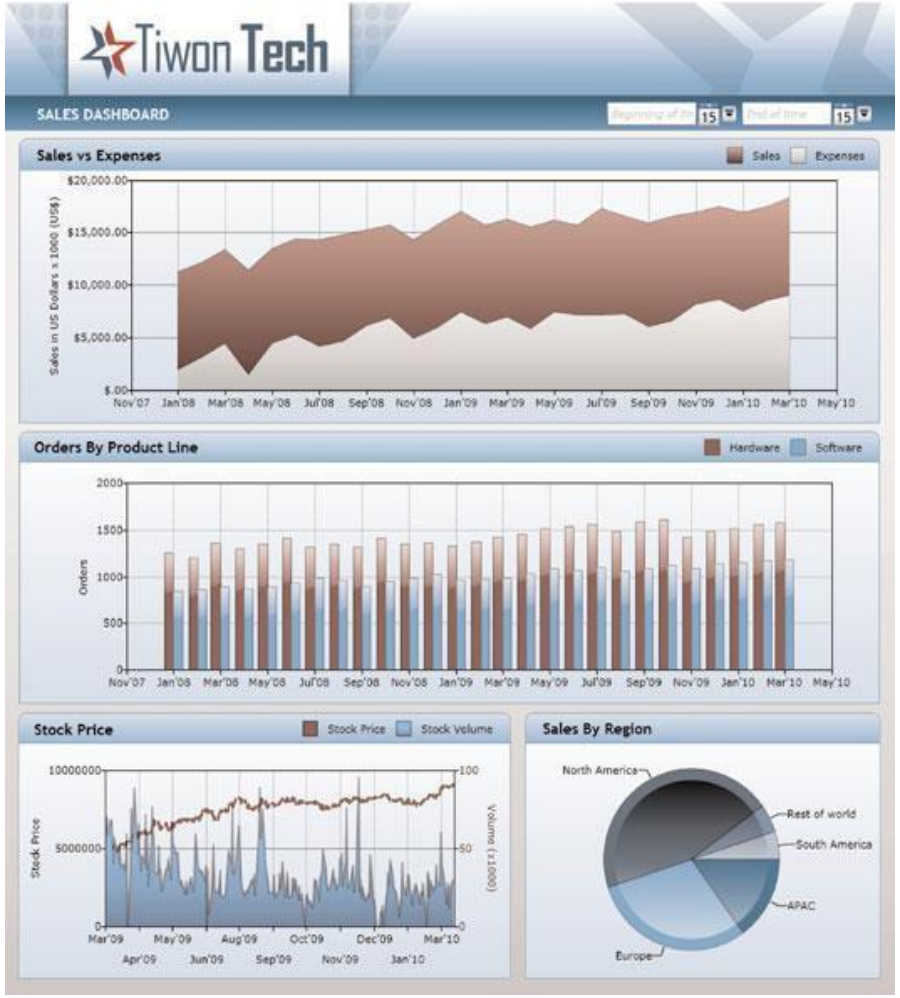
- Serve middle management
- Support non-routine decision making
  - Example: What is impact on production schedule if December sales doubled?
- Often use external information as well from TPS and MIS
  - Example: current stock prices or product prices of competitors

# Types of Information Systems

## 5. Executive Support Systems

- Support senior management
  - Address non-routine decisions
    - Requiring judgment, evaluation, and insight
  - Incorporate data about external events (e.g. new tax laws or competitors) as well as summarized information from internal MIS and DSS
    - Example: Digital dashboard with real-time view of firm's financial performance: working capital, accounts receivable, accounts payable, cash flow, and inventory

# Types of Information Systems




Sales Dashboard

# Types of Information Systems

- **Systems from a constituency perspective**
  - Transaction Processing Systems (TPS):

    supporting operational level employees
  - Management Information Systems And Decision Support Systems (MIS & DSS):

    supporting managers
  - Executive Support Systems (ESS):

    supporting executives

# Types of Information Systems

- **Relationship of systems to one another**
  - TPS: Major source of data for other systems
  - ESS: Recipient of data from lower-level systems
  - Data may be exchanged between systems
  - In reality, most businesses' systems are only loosely integrated (but they are getting better!)

# Types of Information Systems

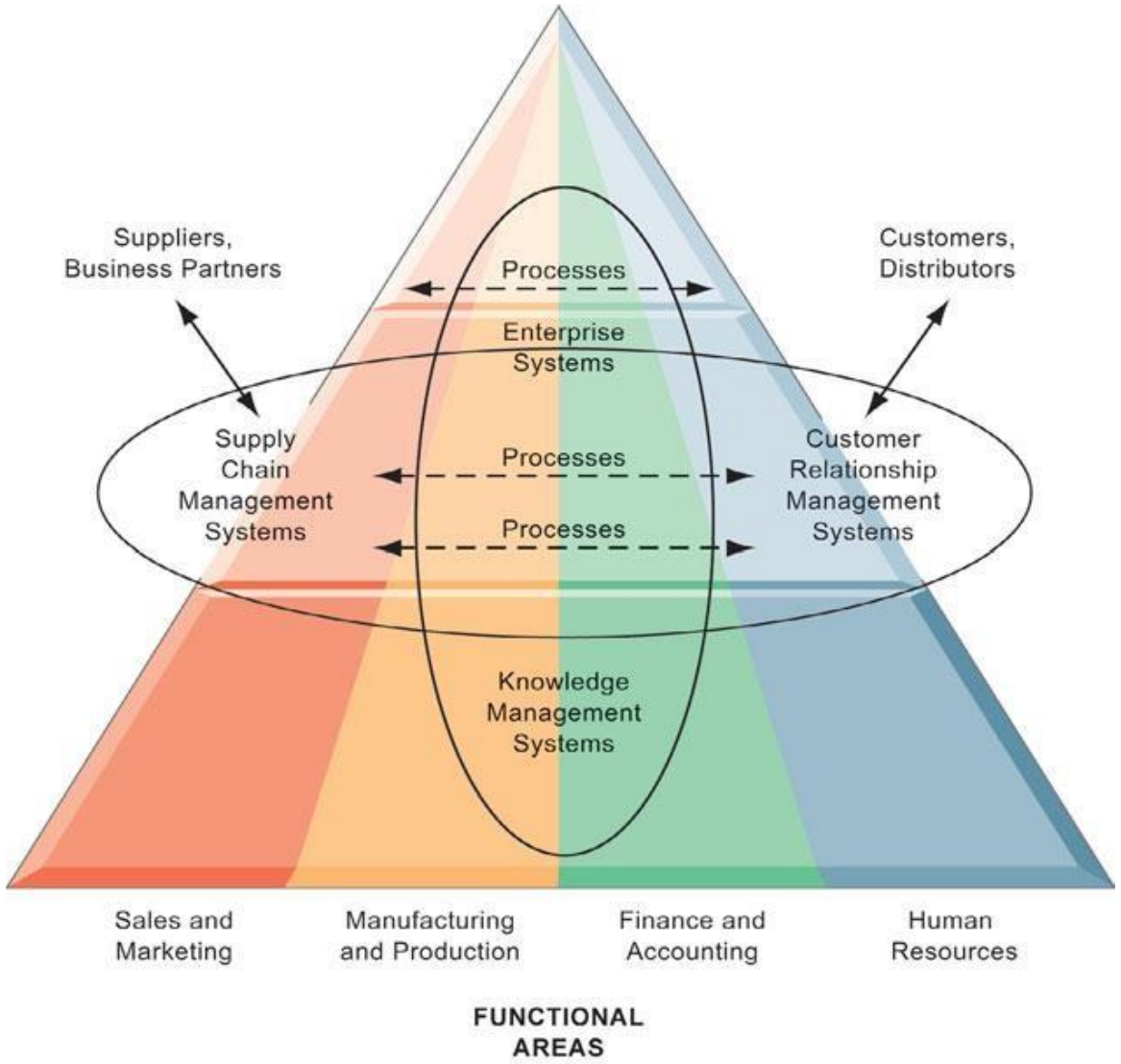
Systems for Linking the Enterprise

- **Enterprise applications**
- Intranets and Extranets

# Types of Information Systems

- **Enterprise applications**
  - System for linking the enterprise
  - Span functional areas
  - Execute business processes across firm
  - Include all levels of management
  - Four major applications:
    - Enterprise systems
    - Supply chain management systems
    - Customer relationship management systems
    - Knowledge management systems

# Types of Information




## ENTERPRISE APPLICATION ARCHITECTURE

Enterprise applications automate processes that span multiple business functions and organizational levels and may extend outside the organization.

# Types of Information Systems

## 1. Enterprise Systems (Enterprise Resource Planning - ERP)

- Collects data from different firm functions and stores data in single central data repository
- Resolves problem of fragmented, redundant data sets and systems
- Enable:
  - Coordination of daily activities
  - Efficient response to customer orders (production, inventory)
  - Provide valuable information for improving management decision making

# Types of Information Systems

## 2. Supply Chain Management (SCM) Systems

- Manage firm's relationships with suppliers
- Share information about
  - Orders, production, inventory levels, delivery of products and services
- Goal:
  - Right amount of products to destination with least amount of time and lowest cost

# Types of Information Systems

## 3. Customer Relationship Management Systems (CRM)

- Provide information to coordinate all of the business processes that deal with customers in sales, marketing, and service to optimize revenue, customer satisfaction, and customer retention

- Integrate firm's customer-related processes and consolidate customer information from multiple communication channels

# Types of Information Systems

## 4. Knowledge Management Systems (KMS)

- Support processes for acquiring, creating, storing, distributing, applying, integrating knowledge
  - How to create,produce,distribute products and services
- Collect internal knowledge and experience within firm and make it available to employees
- Link to external sources of knowledge

# Types of Information Systems

Systems for Linking the Enterprise

- Enterprise applications
- **Intranets and Extranets**

# Types of Information Systems

Alternative tools that increase integration and expedite the flow of information

- Intranets:
  - Internal company Web sites accessible only by employees

- Extranets:
  - Company Web sites accessible externally only to vendors and suppliers
  - Often used to coordinate supply chain

# Types of Information Systems

**E-business**

- Use of digital technology and Internet to drive major business processes

**E-commerce**

- Subset of e-business
- Buying and selling goods and services through Internet

**E-government**

- Using Internet technology to deliver information and services to citizens, employees, and businesses

# Summary

Business Processes and Information Systems
Types of Information Systems

- Systems for Different Management Groups
  - Transaction processing systems
  - Systems for Business intelligence

- Systems for Linking the Enterprise
  - Enterprise applications
  - Intranets and Extranets

- E-business, E-commerce and E-government

# Questions ???



# Thank you…